Lecture-3
Fluid Statics I: Pressure Variation in a Static Fluid

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Learning Objectives & Recap

By the end of this lecture, you should be able to

1. Differentiate fluid **statics versus dynamics**
2. Explain the **pressure variation in a static fluid**
3. Exploit the idea of pressure variation to design a **manometer**

What is fluid statics?

- Statics deals with fluids at rest, or undergoing motion similar to a rigid body, meaning there is **no relative velocity between two points in the fluid**



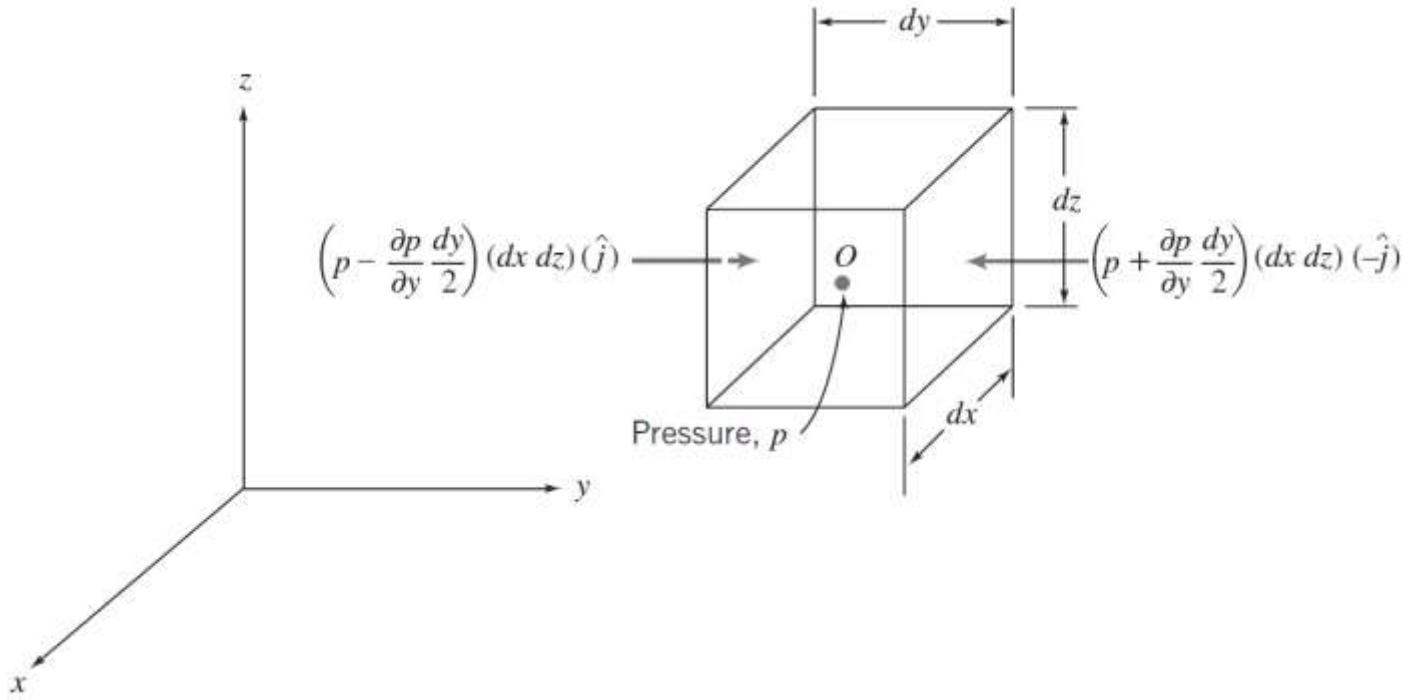




Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.
https://science.howstuffworks.com/transport/engines-equipment/hydraulic.htm

Basic equation of fluid statics

- Our objective is to understand **how the pressure varies** in a static fluid



$$d\vec{F}_S = -\left(\frac{\partial p}{\partial x}\hat{i} + \frac{\partial p}{\partial y}\hat{j} + \frac{\partial p}{\partial z}\hat{k}\right)dx\,dy\,dz$$

$$-\nabla p + \rho\vec{g} = 0$$

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Revisiting the learning objectives

1. Differentiate fluid **statics from dynamics**
2. Explain the **pressure variation in a static fluid**
3. Exploit the idea of pressure variation to design a **manometer**

